

V Muhammed Saad Sabeel

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CAREER OBJECTIVE

To build stuff that matters and make the most of my skills.

EDUCATION

B.E, Electronics and Telecommunication Engineering (ETE)

2023 - 2027

BMS Institute of Technology

CGPA: 9.21/10

PORTFOLIO

[GitHub link ↗](#)

PROJECTS

EVSE Charge Controller Simulator

Jun 2026 - Jul 2026

Built an EVSE charge controller simulator using a FreeRTOS task scheduler to manage concurrent charging states, sensor reading, and safety protocols. Used hardware interrupts and binary semaphores for rapid fault detection, and generated dynamic PWM signals to simulate IEC 61851 communication.

Smart Switching Using IR Remote and Mobile

Jul 2024

Developed a home automation system using a 4-channel relay module for appliance control. Implemented dual control modes: an IR remote for local control and the Blynk IoT platform for mobile access.

IoT Intrusion Detection via Side-Channel Analysis

May 2026 - Present

Built a non-intrusive edge monitoring system that uses hardware power analysis to detect malware, bypassing the need for software logs. Created an I2C data pipeline to sample current consumption every 0.2 seconds, building a dataset to train real-time Edge AI classification models.

QNX Deterministic Security Monitor

Jan 2026 - Feb 2026

Designed a security monitor on a QNX Microkernel using strictly isolated C programs to mitigate hardware side-channel attacks. Developed a high-priority mitigation engine triggered by IPCs to instantly freeze compromised processes mid-instruction and prevent data theft.

Diabreeze: Non-Invasive Biomarker Detection System

Jul 2026 - Present

Built a portable diagnostic system that processes real-time breath acetone telemetry to estimate blood glucose levels. Engineered an ESP32 data pipeline to instantly compute and visualize physiological and environmental metrics.

SKILLS

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|---------|----------------|----------------|
| • C# | • Python | • SQL |
| • ESP32 | • Raspberry Pi | • Keil uVision |
| • Git | • VS Code | • Vivado |

EXTRA CURRICULAR ACTIVITIES

- Webmaster | IEEE Circuits and Systems Society (CASS): Managed and maintained the IEEE CASS student chapter website, ensuring timely updates of announcements, events, and technical content. Collaborated with the organizing team to design and publish event pages.
- AI Model Evaluator | Outlier AI: Contributed 330+ hours of RLHF and SFT tasks to train enterprise LLMs, evaluating model outputs for factual accuracy and logical consistency. Performed quality evaluation of AI-generated outputs across Mathematics and multilingual domains.